

Board color is arithmetic

Chess Math

THE IDEA

Number files a=1 through h=8. A square is dark when file number plus rank is even, and light when the sum is odd. This gives a quick check for bishop routes and knight color changes.

WORKED EXAMPLE

For c6, $3+6=9$, so it is light. For f4, $6+4=10$, so it is dark. A bishop on c6 can never reach f4, even if the board is empty, because the colors differ.

TRY IT BEFORE READING ON

What color is h8, and can a bishop from c3 reach it on an empty board?

ANSWER AND REASONING

h8 is dark because $8+8=16$. c3 is also dark, and h8 lies on its diagonal: both coordinates increase by five. Matching color is necessary but direct diagonal alignment must also be checked.

TAKE THIS INTO YOUR NEXT GAME

Use parity to catch impossible bishop calculations.